CSS Essentials: Colours, Fonts and Boxes

HTML builds the page. CSS makes it beautiful!

Web Dev | Beginner | Ages 9–16

InfiBee Coders

The Big Idea

- HTML is the structure of a web page. CSS is the style: colours, fonts, spacing and layout.
- A CSS rule has a selector and declarations: `selector { property: value; }`.
- Every declaration ends with a semicolon.
- Keep CSS in its own file and link it to every page.

3 Ways to Add CSS

- External file (best): `<link rel="stylesheet" href="style.css">` inside `<head>`.
- Internal: a `<style>` block inside `<head>`.
- Inline: `style="color: red;"` on one element. Use rarely.
- If rules clash, the more specific rule wins, and inline beats the others.

Selectors

- Element: `p { }` styles every paragraph.
- Class: `.card { }` styles anything with `class="card"`. Use classes a lot.
- ID: `#title { }` styles one element with `id="title"`. IDs must be unique.
- Group: `h1, h2 { }` styles both. Descendant: `ul li { }` styles li inside ul.
- State: `a:hover { }` styles a link when the mouse is over it.

Colours, Text and Boxes

- Colours: names (`red`), hex (`#ff6600`) or `rgb(255, 102, 0)`.
- Text: `color`, `font-family`, `font-size`, `font-weight`, `text-align`, `line-height`.
- Box model: content, then `padding`, then `border`, then `margin` on the outside.
- `border: 2px solid #333;` and `border-radius: 12px;` give a border with rounded corners.
- `box-sizing: border-box;` keeps width and height easy to predict.
- Units: `px` is fixed, `%` is relative to the parent, `rem` follows the base font size.

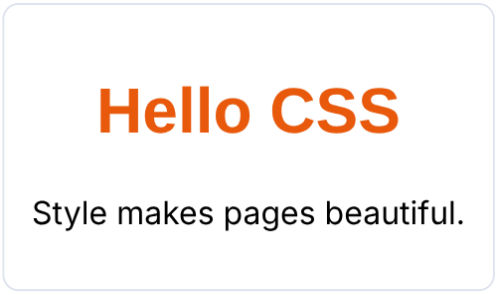
💻 Code Examples

Your first style rule — HTML + CSS

```
<h1>Hello CSS</h1>
<p>Style makes pages beautiful.</p>

h1 {
  color: #e8590c;
  font-family: Arial, sans-serif;
  text-align: center;
}
```

▶ What you see:



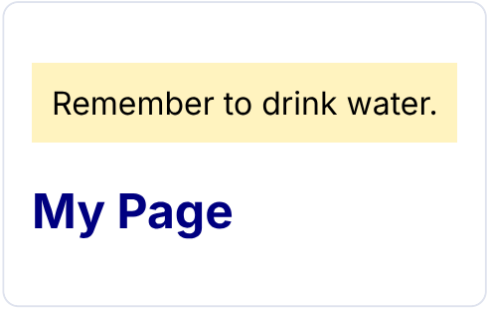
Every h1 heading is orange, centred and uses Arial.

Class and ID — HTML + CSS

```
<p class="note">Remember to drink water.</p>
<h2 id="title">My Page</h2>

.note { background: #fff3bf; padding: 10px; }
#title { color: navy; }
```

▶ What you see:



A class can style many elements. An ID styles just one.

A card using the box model — HTML + CSS

```
<div class="card">
  <h3>Python Club</h3>
  <p>Every Saturday</p>
</div>

.card {
  width: 220px;
  padding: 16px;
  border: 2px solid #333;
  border-radius: 12px;
  background: #fffbea;
}
```

▶ What you see:



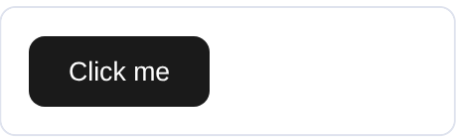
padding is the space inside the border.

A button with hover — HTML + CSS

```
<button>Click me</button>

button {
  background: #1a1a1a;
  color: white;
  padding: 10px 20px;
  border: none;
  border-radius: 8px;
}
button:hover {
  background: #e8590c;
}
```

▶ What you see:



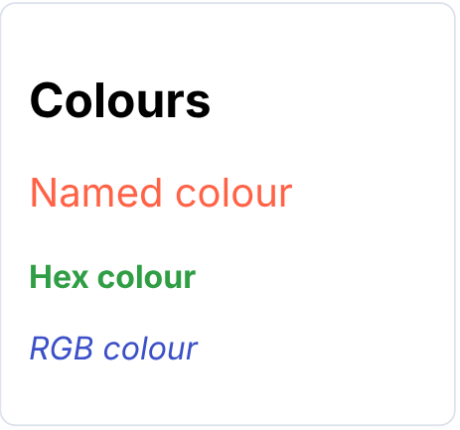
The screenshot shows the normal state. When the mouse moves over it, the button turns orange.

Fonts and colours — HTML + CSS

```
<h2>Colours</h2>
<p class="a">Named colour</p>
<p class="b">Hex colour</p>
<p class="c">RGB colour</p>

.a { color: tomato; font-size: 20px; }
.b { color: #2f9e44; font-weight: bold; }
.c { color: rgb(60, 80, 200); font-style:
italic; }
```

▶ What you see:



You can write colours as names, hex codes or rgb values.

Link the CSS file — HTML

```
<head>
  <link rel="stylesheet" href="style.css">
</head>
```

▶ What happens:
The page now uses every rule in style.css.

📚 Key Terms

CSS
Cascading Style Sheets. The language that styles HTML.

Selector
Says which elements a rule applies to.

Property
What you change, like color.

Value
The setting for the property, like red.

Declaration
One property and value pair.

Class
A reusable name, selected with a dot.

ID
A unique name, selected with a hash.

Padding
Space inside the border.

Margin
Space outside the border.

Pseudo-class
A state of an element, like :hover.

🧮 Quick Reference

```
selector { property: value; }
p { }  .card { }  #title { }
color: #ff6600;
font-size: 18px;
padding: 10px;  margin: 20px auto;
border: 2px solid #333;
border-radius: 12px;
a:hover { color: red; }
```

✏️ Worked Examples

Q: Make all paragraphs blue and 18px.

A: Write p { color: blue; font-size: 18px; }. The p selector targets every paragraph.

Q: Style only some paragraphs.

A: Add class="note" in the HTML to those paragraphs, then write .note { ... } in CSS.

Q: Centre a 300px wide box.

A: Use width: 300px; margin: 0 auto; The auto margins share the space on both sides equally.

🚫 Common Mistakes

- Forgetting a semicolon at the end of a declaration.
- Using `#` for a class or `.` for an ID. Classes use a dot, IDs use a hash.
- Writing `colour` instead of `color`. CSS uses the US spelling.
- Forgetting to link the CSS file, or using the wrong file path.
- Using the same ID on many elements. Use a class instead.

👨‍👩‍👧 Parent Corner

- Ask your child to change one colour and refresh. Instant feedback builds confidence.
- Ask them to find a website they like and describe its colours and fonts.
- Talk about the box model using a gift: the gift is the content, the wrapping is the padding, the box is the border.
- Encourage small experiments. Nothing breaks permanently.
- Ask them to explain what a selector does in their own words.

🎯 Quick Check

1. What does CSS do?

✅ It styles web pages: colours, fonts, spacing and layout.

2. How do you select a class called card?

✅ Use .card in the CSS.

3. What is the difference between padding and margin?

✅ Padding is space inside the border. Margin is space outside it.

4. Which line links a CSS file?

✅ <link rel="stylesheet" href="style.css"> in the head.

5. What does a:hover style?

✅ A link while the mouse is over it.